

Study of Resonances in 1-D Systems by Smooth Exterior Scaling and Complex Scaling

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April 2025

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Quantum Resonances

- Resonance: Metastable quantum states with finite lifetime.
- These states are considered superpositions of bound and continuum states. The different states can be identified using complex scaling, enabling effective identification of resonance states.
- Important in:
 - high-energy physics (like in plasma and x-ray spectroscopy)
 - Electron-molecule collisions relevant to interstellar chemistry
 - understanding DNA damage caused by slow electrons
- Mathematical challenge: non-square-integrable wavefunctions and hence require non-Hermitian treatment.

Non-Hermitian Treatments

- Resonance states are not square integrable.
- Standard Hermitian Hamiltonians yield real eigenvalues only.
Hamiltonians must be analytically continued into the complex plane.
- Enables extraction of complex eigenvalues:

$$E = E_r - i\frac{\Gamma}{2}$$

- Γ : Related to resonance lifetime $\tau = \hbar/\Gamma$.

Principle of Complex Scaling

- Coordinate rotation: $x \rightarrow xe^{i\theta}$, $0 < \theta < \pi/2$.
- Kinetic term scales as $e^{-2i\theta} T$, potential $V(xe^{i\theta})$.
- Bound states remain unaffected; continuum rotates by 2θ .
- Resonance states emerge as isolated points.

Mathematical Formulation

Starting Hamiltonian:

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x).$$

After transformation: $x \rightarrow x' e^{i\theta}$

The Hamiltonian becomes:

$$\hat{H}(\theta) = -\frac{\hbar^2}{2m} e^{-2i\theta} \frac{d^2}{dx'^2} + V(x' e^{i\theta}).$$

Modified Schrödinger equation:

$$\hat{H}(\theta) = T(e^{-2i\theta}) + V(x' e^{i\theta})$$

Solve $\hat{H}(\theta)\phi = E\phi$ for complex E .

Toy Model Potential

- We use the potential:

$$V(x) = \left(1 - 1/\cosh^2(x)\right)e^{-0.05x^2}$$

as mentioned in Ido Gilary's article.

- Supports resonance near $E_r \approx 0.465$, $\Gamma \approx 0.004$.

Eigenvalue Flow: Complex Scaling

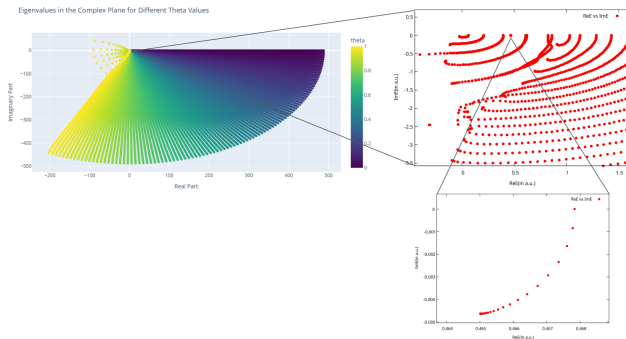


Figure: Eigenvalues under CS vs θ .

Resonance value found at $\approx 0.465 + i0.004$.

SES Transformation

- Define complex path $z = F(x)$ with

$$F'(x) = f(x) = 1 + (e^{i\theta} - 1)g(x),$$

- $g(x)$: Smooth function centered around x_0 , such that $g(x)$ smoothly transitions from 0 (interior) to 1 (exterior).

$$g(x) = 1 + \frac{1}{2} \left[\tanh \left(\frac{x - x_0}{\epsilon} \right) - \tanh \left(\frac{x + x_0}{\epsilon} \right) \right]$$

Effective Complex Absorbing Potential

After SES and rescaling of the wavefunction as $\phi = \psi/\sqrt{f}$, an extra term appears in the schrödinger equation:

$$V_{CAP}(x) = -\frac{\hbar^2}{2m} \left(\frac{-3(f')^2}{4f^4} + \frac{f''}{2f^3} \right).$$

This emerges as an effective complex absorbing potential (CAP) that is mathematically equivalent to the transformation applied to the Hamiltonian outside the interaction region. Its main purpose is to absorb outgoing wavepackets at the boundaries of the computational grid, thereby preventing artificial reflections that can corrupt the calculation of resonance states and scattering processes.

Effective transformed hamiltonian

$$-\frac{\hbar^2}{2m} \left[\frac{1}{f(x)} \frac{d^2}{dx^2} \frac{1}{f(x)} + \left(-\frac{3}{4} \frac{(df/dx)^2}{f(x)^4} + \frac{1}{2} \frac{d^2 f/dx^2}{f(x)^3} \right) + V_F(x) \right] \phi = E\phi$$

or the transformed Schrödinger equation is now read as:

$$-\frac{\hbar^2}{2m} \left[\frac{1}{f(x)} \frac{d^2}{dx^2} \frac{1}{f(x)} + V_{CAP} + V_F(x) \right] \phi = E\phi$$

Hamiltonian Assembly:

$$H_\theta = M T M + V_{CAP} + V_F,$$

where $M = \text{diag}(1/f(x))$, T kinetic, V_F physical potential. Diagonalize for each θ .

Eigenvalue Flow: SES

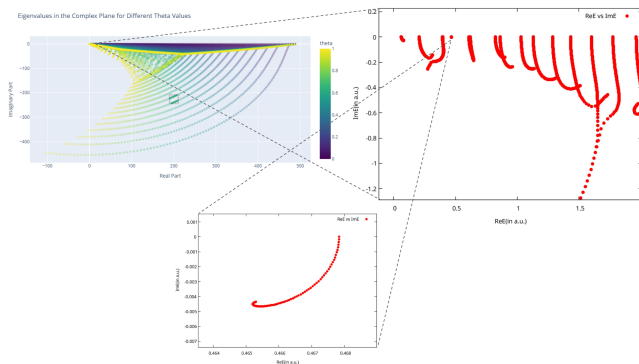


Figure: Eigenvalues under Smooth Exterior Scaling.

Resonance position found around $(0.465, i0.004)$.

Validation Against Literature

- The continuum branch continues to rotate in the complex plane.
- The resonance point remains fixed, the signature of the true metastable state.
- I. Gilary (2012) reports resonance value at : $E_r = 0.463$, $\Gamma = 0.004$ for the toy potential. [2]
- Our values: 0.465 ± 0.002 , 0.004 ± 0.001 .
- Good agreement within numerical error.

Conclusions

- Successfully implemented SES and CS in 1D for finding resonance states.
- Identified resonance at $0.465 + i0.004$.
- Resolved previous discrepancies in the previous SES derivation.

References I

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